AI-ASSISTED CODING

ASSIGNMENT 17.5

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Task 1 –Loan Approval Dataset Cleaning

Use AI to generate preprocessing code for a loan dataset.

Instructions:

- **Handle missing values in loan_amount, credit_score.**
- **Encode loan_status (Approved/Rejected).**
- **Scale numerical features using standard scaling.**
- **Split the dataset into training and testing sets.**

Sample Code:

```
import pandas as pd
data = pd.read_csv("loadapproval.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/amineipad/loan-approval-dataset

Expected Output:

- **A machine-learning-ready loan dataset.**

PROMPT

```python
import pandas as pd
data = pd.read_csv('/content/drive/MyDrive/loanapproval.csv')
print(data.head())
```

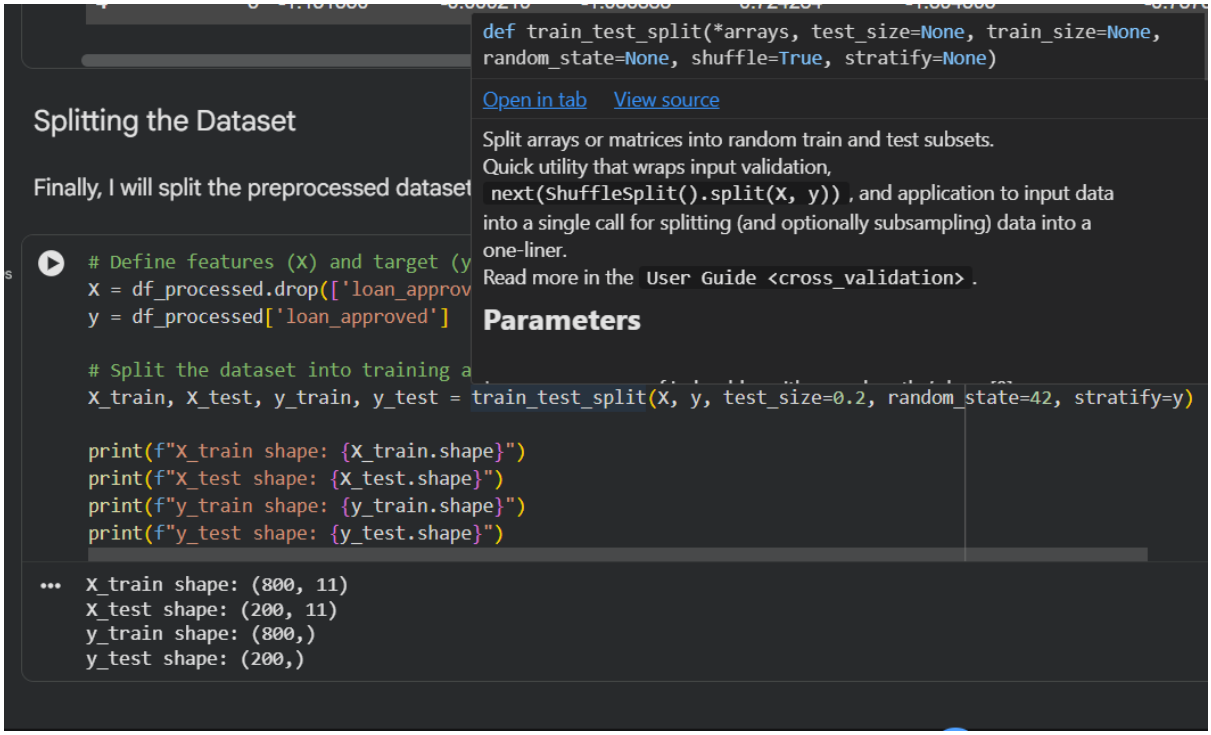
```
   applicant_id  age  gender marital_status  annual_income  loan_amount  \
0             1   59    Male       Divorced         100073         7169
1             2   49    Male        Married         112197        23556
2             3   35    Male       Divorced          84429        27052
3             4   63  Female         Single         124195        11313
4             5   28  Female        Married          81627        13315

   credit_score  num_dependents  existing_loans_count employment_status  \
0           793               1                     1        Unemployed
1           789               0                     2          Employed
2           372               1                     4        Unemployed
3           808               3                     4     Self-employed
4           689               0                     1        Unemployed

   loan_approved
0              1
1              1
2              0
3              1
4              1
```

```python
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
```

```python
# Fill missing values in 'loan_amount' and 'credit_score' with their mean
df_processed['loan_amount'].fillna(df_processed['loan_amount'].mean(), inplace=True)
df_processed['credit_score'].fillna(df_processed['credit_score'].mean(), inplace=True)

print("Missing values after handling:")
print(df_processed[['loan_amount', 'credit_score']].isnull().sum())
```

```
Missing values after handling:
loan_amount     0
credit_score    0
dtype: int64
/tmp/ipykernel_5847/1259966653.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate obj

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplac


  df_processed['loan_amount'].fillna(df_processed['loan_amount'].mean(), inplace=True)
/tmp/ipykernel_5847/1259966653.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate obj

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplac


  df_processed['credit_score'].fillna(df_processed['credit_score'].mean(), inplace=True)
```

Encoding Categorical Features



CODE

OUTPUT

Task 2 – Social Media Profiles

Preprocess a social media engagement dataset using AI assistance.

Instructions:

- **Remove duplicate user profiles.**
- **Handle missing values**
- **Encode categorical columns**

Dataset link:

https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles

Expected Output:

- **A cleaned dataset for engagement prediction.**

```
import pandas as pd
import re

# Load the dataset
data = pd.read_csv('/content/drive/MyDrive/SocialMediaTop100.csv', encoding='latin1')
```

Data Inspection

```
# Display the first few rows of the dataset
display(data.head())

# Get a concise summary of the DataFrame
print(data.info())

# Check for the number of unique user profiles (assuming each row is a profile, or a unique identifier
# For now, let's check for duplicate rows in the entire dataset
print(f"Number of duplicate rows: {data.duplicated().sum()}")

# Check for missing values
print("\nMissing values per column:")
print(data.isnull().sum())
```

	applicant_id	age	gender	marital_status	annual_income	loan_amount	credit_score	num_dependents
0	1	59	Male	Divorced	100073	7169	793	1
1	2	49	Male	Married	112197	23556	789	0
2	3	35	Male	Divorced	84429	27052	372	1
3	4	63	Female	Single	124195	11313	808	3
4	5	28	Female	Married	81627	13315	689	0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 11 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   applicant_id          1000 non-null   int64
 1   age                   1000 non-null   int64
 2   gender                1000 non-null   object
 3   marital_status        1000 non-null   object
 4   annual_income         1000 non-null   int64
 5   loan_amount           1000 non-null   int64
 6   credit_score          1000 non-null   int64
 7   num_dependents        1000 non-null   int64
 8   existing_loans_count  1000 non-null   int64
 9   employment_status     1000 non-null   object
 10  loan_approved         1000 non-null   int64
dtypes: int64(8), object(3)
memory usage: 86.1+ KB
None
Number of duplicate rows: 0
```

```
Missing values per column:
applicant_id            0
age                     0
gender                  0
marital_status          0
annual_income           0
loan_amount             0
credit_score            0
num_dependents          0
existing_loans_count    0
employment_status       0
loan_approved           0
dtype: int64
```

Remove Duplicate User Profiles

```
# Remove duplicate rows
data_processed = data.drop_duplicates().copy()

print(f"Shape of data before removing duplicates: {data.shape}")
print(f"Shape of data after removing duplicates: {data_processed.shape}")
```

```
Shape of data before removing duplicates: (1000, 11)
Shape of data after removing duplicates: (1000, 11)
```

Encode Categorical Columns

```
# Identify categorical columns for encoding (object dtype)
categorical_cols = data_processed.select_dtypes(include=['object']).columns

# Apply one-hot encoding using pd.get_dummies
data_processed = pd.get_dummies(data_processed, columns=categorical_cols, drop_first=True)

print("\nDataFrame after one-hot encoding:")
display(data_processed.head())
print(f"New shape of the dataset: {data_processed.shape}")
```

```
DataFrame after one-hot encoding:
```

	applicant_id	age	annual_income	loan_amount	credit_score	num_dependents	existing_loans_count	lo
0	1	59	100073	7169	793	1	1	
1	2	49	112197	23556	789	0	2	
2	3	35	84429	27052	372	1	4	
3	4	63	124195	11313	808	3	4	
4	5	28	81627	13315	689	0	1	

```
New shape of the dataset: (1000, 13)
```

Task 3 – Weather Dataset Feature Engineering

Task:

Use AI to preprocess a weather dataset for forecasting.

Instructions:

- **Convert date column into datetime format.**
- **Handle missing values in temperature, humidity.**

• Create new features (season, week_number).

• Normalize rainfall values.

Dataset link:

https://www.kaggle.com/code/alsamaualalhinai/weather-dataset/input

Expected Output:

• A processed dataset ready for forecasting models.

Task 4 – Movie Ratings Dataset Preparation

Task:

Clean and preprocess a movie ratings dataset using AI-generated scripts.

Instructions:

• Handle missing values in rating, genre.

• Encode categorical variables like genre, language.

• Remove duplicate movie entries.

• Scale rating values between 0 and 1.

Dataset link:

https://www.kaggle.com/datasets/PromptCloudHQ/imdb-data

Expected Output:

• A dataset suitable for recommendation systems.

```
data = pd.read_csv('/content/drive/MyDrive/IMDB-Movie-Data.csv')
print(data.head)
```

```
<bound method NDFrame.head of      Rank                    Title                     Genre  \
0       1  Guardians of the Galaxy   Action,Adventure,Sci-Fi
1       2               Prometheus  Adventure,Mystery,Sci-Fi
2       3                    Split           Horror,Thriller
3       4                     Sing   Animation,Comedy,Family
4       5            Suicide Squad  Action,Adventure,Fantasy
..    ...                      ...                       ...
995   996     Secret in Their Eyes       Crime,Drama,Mystery
996   997          Hostel: Part II                    Horror
997   998   Step Up 2: The Streets       Drama,Music,Romance
998   999             Search Party          Adventure,Comedy
999  1000               Nine Lives     Comedy,Family,Fantasy

                                        Description              Director  \
0    A group of intergalactic criminals are forced ...            James Gunn
1    Following clues to the origin of mankind, a te...          Ridley Scott
2    Three girls are kidnapped by a man with a diag...    M. Night Shyamalan
3    In a city of humanoid animals, a hustling thea...  Christophe Lourdelet
4    A secret government agency recruits some of th...            David Ayer
..                                                 ...                   ...
995  A tight-knit team of rising investigators, alo...             Billy Ray
996  Three American college students studying abroa...              Eli Roth
997  Romantic sparks occur between two dance studen...            Jon M. Chu
998  A pair of friends embark on a mission to reuni...         Scott Armstrong
```

```
0    Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S...  2014
1    Noomi Rapace, Logan Marshall-Green, Michael Fa...  2012
2    James McAvoy, Anya Taylor-Joy, Haley Lu Richar...  2016
3    Matthew McConaughey,Reese Witherspoon, Seth Ma...  2016
4    Will Smith, Jared Leto, Margot Robbie, Viola D...  2016
..                                                ...   ...
995  Chiwetel Ejiofor, Nicole Kidman, Julia Roberts...  2015
996  Lauren German, Heather Matarazzo, Bijou Philli...  2007
997  Robert Hoffman, Briana Evigan, Cassie Ventura,...  2008
998  Adam Pally, T.J. Miller, Thomas Middleditch,Sh...  2014
999  Kevin Spacey, Jennifer Garner, Robbie Amell,Ch...  2016

     Runtime (Minutes)  Rating   Votes  Revenue (Millions)  Metascore
0                  121     8.1  757074              333.13       76.0
1                  124     7.0  485820              126.46       65.0
2                  117     7.3  157606              138.12       62.0
3                  108     7.2   60545              270.32       59.0
4                  123     6.2  393727              325.02       40.0
..                 ...     ...     ...                 ...        ...
995                111     6.2   27585                 NaN       45.0
996                 94     5.5   73152               17.54       46.0
997                 98     6.2   70699               58.01       50.0
998                 93     5.6    4881                 NaN       22.0
999                 87     5.3   12435               19.64       11.0

[1000 rows x 12 columns]>
```

Data Inspection

Data Inspection

```python
# Create a copy of the original DataFrame to work with
df_processed = data.copy()

# Display basic information about the dataset
print("Dataset Info:")
df_processed.info()

# Check for missing values
print("\nMissing values before handling:")
print(df_processed.isnull().sum())

# Check for duplicate movie entries
print(f"\nNumber of duplicate rows before handling: {df_processed.duplicated().sum()}")
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Rank                1000 non-null   int64
 1   Title               1000 non-null   object
 2   Genre               1000 non-null   object
 3   Description         1000 non-null   object
 4   Director            1000 non-null   object
 5   Actors              1000 non-null   object
 6   Year                1000 non-null   int64
```

```
Description               0
Director                  0
Actors                    0
Year                      0
Runtime (Minutes)         0
Rating                    0
Votes                     0
Revenue (Millions)      128
Metascore                64
dtype: int64

Number of duplicate rows before handling: 0
```

Handle Missing Values

I will fill missing numerical values with the median (for 'Revenue (Millions)' and 'Metascore') and categorical string values with 'Unknown' (for 'Genre', 'Director', and 'Actors').

```python
# Fill missing 'Revenue (Millions)' and 'Metascore' with their median
df_processed['Revenue (Millions)'].fillna(df_processed['Revenue (Millions)'].median(), inplace=True)
df_processed['Metascore'].fillna(df_processed['Metascore'].median(), inplace=True)

# Fill missing 'Genre', 'Director', 'Actors' with 'Unknown'
df_processed['Genre'].fillna('Unknown', inplace=True)
df_processed['Director'].fillna('Unknown', inplace=True)
df_processed['Actors'].fillna('Unknown', inplace=True)

print("Missing values after handling:")
```

```python
print(df_processed.isnull().sum())
```

```
Missing values after handling:
Rank                  0
Title                 0
Genre                 0
Description           0
Director              0
Actors                0
Year                  0
Runtime (Minutes)     0
Rating                0
Votes                 0
Revenue (Millions)    0
Metascore             0
dtype: int64
/tmp/ipykernel_5847/2603282821.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate obj

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplac


  df_processed['Revenue (Millions)'].fillna(df_processed['Revenue (Millions)'].median(), inplace=True)
/tmp/ipykernel_5847/2603282821.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate obj

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplac


  df_processed['Metascore'].fillna(df_processed['Metascore'].median(), inplace=True)
/tmp/ipykernel_5847/2603282821.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate obj
```

Gemini

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I will remove any exact duplicate rows from the dataset.

[14]

```python
# Remove duplicate rows
df_processed.drop_duplicates(inplace=True)

print(f"Number of duplicate rows after handling: {df_processed.duplicated().sum()}")
print(f"Shape of DataFrame after removing duplicates: {df_processed.shape}")
```

```
Number of duplicate rows after handling: 0
Shape of DataFrame after removing duplicates: (1000, 12)
```

Encode Categorical Variables (Genre)

The 'Genre' column contains multiple genres per movie. I will convert this into a set of binary (one-hot encoded) columns, one for each unique genre.

[15]

```python
# Extract all unique genres
all_genres = set()
for genres_list in df_processed['Genre'].str.split(','):
    for genre in genres_list:
        all_genres.add(genre.strip())

# Create binary columns for each genre
for genre in all_genres:
    df_processed[f'Genre_{genre}'] = df_processed['Genre'].apply(lambda x: 1 if genre in x else 0)
```

Scale Rating Values

I will scale the 'Rating' column to values between 0 and 1 using `MinMaxScaler`.

[16]
0s

```python
from sklearn.preprocessing import MinMaxScaler

# Initialize MinMaxScaler
scaler = MinMaxScaler()

# Reshape the 'Rating' column for scaling (required for single feature)
df_processed['Rating_scaled'] = scaler.fit_transform(df_processed[['Rating']])

# Optionally, drop the original 'Rating' column if only the scaled version is needed
# df_processed.drop('Rating', axis=1, inplace=True)

print("DataFrame with scaled 'Rating' column:")
display(df_processed[['Rating', 'Rating_scaled']].head())
```

```
DataFrame with scaled 'Rating' column:
```

	Rating	Rating_scaled
0	8.1	0.873239
1	7.0	0.718310
2	7.3	0.760563
3	7.2	0.746479
4	6.2	0.605634